METHOD 12 / A COMPLETE BEGINNER GUIDE

Lagged cross-correlation

Does today's move in series X line up with tomorrow's move in series Y? Lagged cross-correlation measures Pearson correlation between x_t and y_{t+k} for chosen lag k.

It is a **screening** tool, not proof that X forecasts Y profitably. Teaching examples only; experiments not started.

Read it in three parts

First: Idea, percents, series. **Next:** Notation, formula, hand calc. **Last:** FX use, limits, statsmodels CCF check.

Pages	What you will learn
2	Idea; is rho a percent?
3	FX percent example
4	Which series and lag sign
5	Role in time-series screening
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Prepared for the folder: 12 - Lagged Cross-Correlation
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02 / THE BASIC IDEA

What does lagged cross-correlation do?

Same-period correlation asks about x_t and y_t together. Lagged cross-correlation shifts one series forward by k periods and asks whether earlier X moves align with later Y moves.

```
rho_{X->Y}(k) = Corr( x_t, y_{t+k} )
```

Is this a percent?

No. x_t and y_t are returns (often decimal form). rho is on [-1, +1]. A lag k is an integer count of bars, not a percent.

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

```
Simple: 1.1110/1.1000 - 1 = 0.01 (1.00%)
Log: ln(1.1110/1.1000) = about 0.009950 (~0.995%)
```

Build return series first. Then align x_t with y_{t+k}. Mixing quote conventions can flip signs; keep one convention per study.

Lagged cross-correlation uses the same return units as same-day correlation. A rho of 0.40 is not 40% return and not 40% forecast accuracy.

After you compute returns, apply the lag before any optional standardisation step so the time index stays transparent in experiment logs.

04 / DATA RULES

Which series and which lag?

Returns, equal spacing

Use log or simple returns consistently. Each lag k drops k rows at the end of the alignment.

Direction label X to Y

rho_{X->Y}(k) uses X at t and Y at t+k. rho_{Y->X}(k) swaps roles. They need not match.

Not a trading back-test

A high lagged rho can be spurious with many lags tried or with overlapping macro shocks.

05 / CONTEXT

Research and screening role

Cross-correlation at lags is classical time-series diagnostics (Box-Jenkins tradition). In FX research it often appears as a first pass before Granger tests (folder 13) or VAR models.

What we claim

This guide teaches definition and honest limits. We do not cite invented EURUSD lead-lag results. Experiments not started.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
k	Lag in bars (k=1 is one bar ahead for Y).
x_t, y_t	Return at time t.
rho_{X->Y}(k)	Lagged cross-correlation at lag k.
n_k	Number of paired rows after alignment (n - k).

Positive k means Y is shifted forward relative to X in the pairing described on page 7. If your software uses the opposite sign convention, flip the label when you report results.

n_k shrinks as k grows because the first k rows of Y (or last k rows of X) have no partner. Record n_k beside every rho you publish.

07 / ALIGNMENT

The main formula

```
For k >= 1:
pairs: (x_1, y_{1+k}), (x_2, y_{2+k}), ...
rho_{X->Y}(k) = Pearson correlation of those pairs
```

Apply the usual Pearson steps from folder 01 on the aligned subsample. k = 0 is same-bar correlation (different question).

Write the lag explicitly in the report title, for example EURUSD leads GBPUSD by one daily bar, so readers know which direction you tested.

If you reverse direction, you must rebuild the pairs; swapping labels without re-aligning rows is a common spreadsheet error.

Sign of lag

If rho_{X->Y}(1) is large and positive, x today tends to move with y tomorrow in a linear sense. That is not proof X causes Y economically.

Negative k or reverse ordering defines the opposite direction. Plot the cross-correlation function (CCF) across k to see several lags at once.

09 / WORKED EXAMPLE

Calculate by hand: k = 1

Five made-up daily returns (decimals):

t	x_t	y_t
1	0.010	0.015
2	0.020	0.040
3	0.030	0.050
4	0.040	0.070
5	0.050	0.090

For k = 1, pair x at t = 1..4 with y at t = 2..5.

```
Aligned x: [0.010, 0.020, 0.030, 0.040]
Aligned y: [0.040, 0.050, 0.070, 0.090]
rho_{X->Y}(1) = about 0.9898 in this toy list
```

This teaching list is almost perfectly aligned at lag 1. Real FX series rarely look this clean; treat high rho as a prompt for deeper validation, not as proof of edge.

10 / CCF HABIT

Screen several lags

Compute rho(k) for k = 0, 1, ..., K_max. Pre-specify K_max before you look. Trying many lags and reporting only the best is data mining.

FX majors note

Shared session opens can create microstructure lag patterns that differ from daily bars. Match bar size to the question you wrote down first.

11 / WHAT THIS CANNOT PROVE

Limits

Not forecast proof

Correlation at a lag does not include costs, stability, or out-of-sample validation.

Autocorrelation contamination

If each series is highly autocorrelated, lagged cross-correlation can partly reflect own-lag structure. Consider pre-whitening or VAR/Granger paths for deeper tests.

Versus Granger (folder 13)

Granger tests predictability in a regression with multiple lags. Lagged cross-correlation is a lighter single-lag Pearson screen.

Software check

```
import numpy as np
from statsmodels.tsa.stattools import ccf
x = np.array([...])
y = np.array([...])
# ccf(x, y, adjusted=False) gives cross-correlations
# Read docs for normalization and lag order
```

Confirm whether the function returns corr(x_t, y_{t-k}) or corr(x_t, y_{t+k}). Package conventions differ; map explicitly to rho_{X->Y}(k) in this guide.

Check your understanding

Is rho_{X->Y}(k) a percent? No. **Does k=1 prove X forecasts Y?** No, it is a linear screen at one lag. **Should I try many k values?** Pre-set K_max and note multiple-test risk.

Plot the CCF for context, but do not cherry-pick the lag with the largest absolute value without a pre-registration rule.

Sources and further reading

Is rho a percent? No. **Does high rho at k=1 prove tradeable lead?** No. **Should I scan many lags?** Only with a pre-set max and multiple-test caution.

[1] statsmodels.tsa.stattools.ccf documentation.
CCF implementation and lag convention.
Open source

[2] Folder 01 Pearson Correlation guide.
Pearson on aligned pairs.
Open source